

Conformity Effects on Public Opinion in the Face of Looming Social Pressure*

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Abstract

A high-profile shift in social norms in recent decades has been a movement from support for hierarchical structures defining relationships between social groups to rejection of hierarchy in favor of egalitarian values. In the process of such norm displacement, the individuals that comprise a society should be expected to have the newly socially appropriate attitude reinforced through social interactions. Though this process is widely theorized and discussed, it remains difficult to examine empirically because most of the reinforcement takes place in everyday interpersonal interactions, many of which are private. This paper presents a laboratory experiment in which individuals learn that the distribution of opinions on a set of survey questions among a proximate group of peers is opposed to broader egalitarian norms, and that their opinions on those same survey questions will be made public to that peer group. Results indicate that this form of social pressure is at least weakly predictive in moving participants away from egalitarian attitude expressions toward the opinion of the peer group.

Keywords conformity effects, social norms, social pressure, egalitarian norms, political values

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1. Introduction

Norms of individual equality and equality between social groups are staples of democratic political culture. Following long histories of political, economic, and social systems based on gender, racial, religious, and other forms of hierarchy, principles of equal protection of citizens' rights, regardless of identity group, are now codified into law in virtually all modern participatory democracies. Furthermore, the law comports with mass opinion, with support for egalitarianism having achieved the status of a social norm (Feldman and Steenbergen 2001). Across democratic societies—particularly those in the West (Schwarz 2004, 2006)—citizens consistently exhibit high levels of support for the idea that all people should be treated as political and social equals, without regard to group identity characteristics such as gender, race, religion, or socioeconomic status. Based on public opinion measures, support for principles of political and social equality seem to be a pervasive feature of modern democratic polities (Schwarz 2006).

General popular acceptance of egalitarian norms does have limits, however (McClosky and Zaller 1984; Verba and Orren 1985), and those limits often reveal themselves when individuals contemplate intergroup interactions (Gilens 1995; Kteily, Sheehy-Skeffington, and Ho 2017; Lucas and Kteily 2018; Mann and Kawakami 2012). Even genuine support for abstract notions of equality tends to mask a patchwork of compromises that individuals are willing to tolerate in certain situations, as well as mere individual variation in egalitarian values (Schwarz 2006). Additionally, though various forms of inequality persist in modern democracies, citizenries often show limited support for government action to address them (Feldman and Steenbergen 2001; McClosky and Zaller 1984; Van Heuvelen 2017). Even more important, however, the expressed levels of support for certain egalitarian principles may not be genuine at all (Johnson and Van De Vijver 2002).

Because citizens in democratic societies generally believe that egalitarian principles are widely held among their fellow citizens, publicly expressing inegalitarian attitudes may carry the possibility of embarrassment or social sanction, a threat that was likely reinforced via repeated social interactions over time. In the end, among a possibly substantial subset of a given citizenry, self-reports of adherence to egalitarian principles may reflect conformity to social rules that govern appropriate behavior rather than truly held beliefs.

Though it is plausible to believe that this sort of social desirability bias would serve to inflate expressed levels of public support for egalitarian principles, little is known about how the desire to give the appropriate social response operates at the level of interpersonal interactions. That is, within a broader social climate that evokes a strong norm of egalitarianism, we know little about how everyday interpersonal communications serve to perpetuate both the norm of acceptable behavior and variation in individuals' actual attitudes. Empirical studies of social conformity with respect to general, arbitrary norms established through social interactions have a long history in social psychology (e.g., Asch 1951, 1952, 1956; Deutsch and Gerard 1955; Sherif 1935, 1936). However, with respect to political and social attitudes, empirical examinations of conformity at the interpersonal level remain somewhat rare (but see Levitan and Verhulst 2016; Carlson and Settle 2016; Mallinson and Hatemi 2018). Therefore, though we have strong reasons to believe that expressed public support for egalitarian principles on surveys are inflated, we know little about how this norm structures, and is structured by, interactions between individuals.

Like previous recent examinations of the interpersonal dynamics of social pressure to conform (Levitan and Verhulst 2016; Carlson and Settle 2016; Mallinson and Hatemi 2018), the present study uses a laboratory setting to investigate social conformity with respect to deeply-held values: in this case, adherence to egalitarian principles. I present a research design that sidesteps the question of whether expressed opinions about egalitarianism represent respondents' truly held beliefs and

rather focuses attention on the dynamics of social influence with respect to baseline attitudinal measures gathered in an isolated setting. In other words, isolation itself is not expected to completely eliminate social desirability concerns—indeed, social pressure stemming from researcher effects may well be present—but it is expected to provide a contrast with which to compare direct social pressure exerted in peer-group surveillance settings.

The paper proceeds as follows. In the following section I provide a literature review that leads to a specific prediction about conformity to group opinion. In the subsequent section, I lay out an experimental design aimed at testing the difference in expressed attitudes between settings in which individuals do and do not learn about the distribution of opinion from peers with whom they will supposedly interact. The final two sections present results and offer concluding remarks.

II. Background and Predictions

Here I set up the description of a laboratory study on social pressure effects by explaining the circumstances under which we should expect conformity with respect to attitudes. Specifically, I note the distinct nature of attitudes with respect to egalitarian norms as opposed to the types of arbitrary tasks examined in traditional social psychological approaches, and as opposed to the types of attitudinal questions used in recent laboratory studies of conformity with respect to political opinions. I note the ways in which social influences are expected to operate on political values, leading into a specific prediction about the expected behavior in the current study.

1. Egalitarian Norms in Modern Democracies

Support for egalitarianism, as opposed to support for hierarchical social structures, has seen a vast uptick in modern multicultural democracies, particularly in the West (Schwarz 2006). One who expresses support for egalitarianism is communicating a desire for equality between individuals or groups, and this may take the form of equality of opportunity, government policy to engineer more equal outcomes, or interpersonal interactions characterized by equality of social status between the parties (Arneson 2013). Examining trends with respect to, for instance, racial attitudes in the United States (e.g., Schuman et al. 1998), or attitudes about gender roles (e.g., Inglehart and Norris 2003) or the acceptability of homosexuality (e.g., Roberts 2019) across the world, the dominant tendency, by far, is movement toward acceptance and equality between social groups. Egalitarianism, it seems, has reached the status of a social norm in modern democratic culture (Feldman and Steenbergen 2001).

To be sure, however, having achieved the status of a norm implies widespread knowledge of the distribution of expressed opinions. In other words, for an attitude to be a social norm requires mass awareness of the socially appropriate expression. And this awareness in turn leads to some level of ambiguity about whether the observed changes in public opinion trends represent true changes in beliefs, or mere norm adherence (Johnson and Van De Vijver 2002). In reality, changes in aggregate opinion distributions over long spans of time are likely to represent a mixture of generational replacement, real attitude change, and both conscious and sub-conscious masking of truly held attitudes in favor of a norm-consistent response. Precise estimates of the extent to which each of these phenomena is represented in an aggregate opinion trends are elusive.

2. Social Influence and Political Values

Regardless of the degree to which changes in expressed attitudes regarding egalitarian norms represent real attitude change as opposed to mere norm adherence, the mass public nonetheless internalized the change about which opinions are deemed socially acceptable. Scholars are indeed quite familiar with the notion that, with respect to the salient social issues of the day, the citizens that make up aggregate public opinion distributions tend to be in tune with the norms of behavior that regulate the range of opinions deemed to be socially acceptable (e.g., Crowne and Marlowe 1960; Kuran 1995; Mendelberg 2001; Noelle-Neumann 1984). Furthermore, this internalization process likely resulted from embeddedness in a socio-cultural context that structures interpersonal social interactions. Learning over time through everyday interactions ultimately steers individuals in the same social milieu toward a shared understanding of the acceptability of certain opinions and behaviors (Hardin and Higgins 1996). The importance of interpersonal interactions in exerting social pressure is widely recognized, but notoriously difficult to study in real-world settings, particularly with respect to deeply held values.

Large bodies of scholarship in political science have been concerned with interpersonal influences over political attitudes and behaviors, but such research has largely dealt with homophily, learning, and information exchange rather than normative compliance. For instance, social networks structure the nature and types of interactions that individuals experience, and those interactions in turn can lead to actual changes in political attitudes (Huckfeldt and Sprague 1995; Mutz 2002, 2006; Mutz and Mondak 2006; Sinclair 2012). Interpersonal interactions also serve as channels of information curation by opinion leaders (Katz 1957; Katz and Lazarsfeld 1955; Lazarsfeld, Berelson, and Gaudet 1948) and information exchange between peers (Ahn, Huckfeldt, and Ryan 2010; Barabas 2004; Bøggild, Aarøe, and Petersen 2021; Carlson 2019). But even without any exchange of factual in-

formation, proximate others in one's social setting still provide benchmarks for the range of behaviors and opinions that are considered socially appropriate (Deutsch and Gerard 1955; Festinger 1954; Kelley 1952).

More relevant methodologically for the present study, a long tradition in social psychology examines the interpersonal dynamics of social pressure in laboratory settings (Sherif 1935, 1936; Asch 1951, 1952; Bond and Smith 1996). Using confederates and an optical illusion known as the autokinetic effect (Adams 1912), Sherif (1935, 1936) demonstrated that when human subjects are asked to express judgments about ambiguous phenomena without a clear right or wrong answer, there is a strong tendency to defer to the opinion of the majority. Even more famously, Asch (1951, 1952) extended the work of Sherif by examining interpersonal social pressure to conform in situations in which the objectively correct response was unambiguous. Specifically, subjects in Asch's most famous studies were asked to examine the lengths of lines drawn on index cards and express which lines appeared to be the same length. The famous twist was that the real subject followed a group of confederates who all provided the incorrect judgment, setting up a choice between publicly expressing a contrarian view or providing the objectively correct response. Across a variety of versions of this basic design, Asch (1951, 1952) found that over one-third of subjects were willing to give the incorrect response. Overall, laboratory experiments on the interpersonal dynamics underlying conformity point to a strong tendency for people who are compelled to make judgments publicly to comply with expectations about the appropriate social response.

While the research programs of Asch and Sherif provide a useful framework for examining social conformity driven by interpersonal interactions, they seem particularly strong with respect to tasks that have no social significance for the subjects outside of the lab. Only recently have political scientists employed experiments leveraging interpersonal interactions within this general framework to study changes in expressed political attitudes in the face of social pressure (Leviton and Verhulst

2016; Carlson and Settle 2016; Mallinson and Hatemi 2018). But because these studies focus on attitudes with respect to identity concerns or political issues that cut along a traditional liberal-conservative ideological framing, it is not completely clear how they relate to the infamous Asch results. Specifically, because the attitudinal questions asked in these studies do not tap issues for which there exists a clear normative push to answer in one particular direction, they may indeed be relatively malleable. The present study extends this recent work by examining the propensity to conform to the opinions of others when there exists a known correct social response.

3. Hypothesis

The Asch and Sherif research programs, as well as contemporary laboratory studies in political science (Levitan and Verhulst 2016; Carlson and Settle 2016; Mallinson and Hatemi 2018), all lead to the general implication that one should expect interpersonal social pressure to lead to a tendency to alter judgments in the face of opposition. Like ambiguous decision-making settings such as responding to optical illusions, expressions of political attitudes may be subject to social pressure effects (or not) because there is not a clear correct answer from the subject's perspective. However, when those attitudinal questions tap notions of egalitarian norms, the situation may come to more closely resemble unambiguous decision-making settings such as matching the lengths of lines drawn on cards in that there is a clearly correct social normative response. Given the strong findings from previous research that at least a substantial minority is willing to provide objectively incorrect responses, I offer the following prediction:

Hypothesis: Individuals who learn that the distribution of attitudes in their immediate

peer group is counter to broader social norms and are compelled to provide their own attitudinal responses will be more likely to exhibit counter-normative attitudes than individuals who answer in isolation.

In other words, if a clear picture of the distribution of expressed attitudes in the population can be established among individuals compelled to express those attitudes in isolation, the prediction is that individuals can be lured away from that population distribution if they are compelled to provide their own attitudes publicly and in the face of an opposed group of peers.

III. The Social Pressure Experiment

I now present an experimental design that was employed to test the hypothesis put forth in the previous section. The study took place from September to December 2011 and used as subjects undergraduate political science students from the University of Illinois at Urbana-Champaign. The objective was to use a between-subjects design to examine the effect on the propensity to conform of a compound treatment consisting of (1) informing subjects about the opinion distribution of another participant in the study, and (2) informing subjects that their own opinions will be revealed to that same participant at some point in the future. In other words, the design uses a survey battery to tap the attitudes of one experimental group in isolation, with no information about or social interactions with other participants. Then it assesses the attitudes of another group of subjects who are provided with the responses of another participant and are told that their own responses will be revealed to that participant during a face-to-face interaction. In

other words, using a between-subjects design, a control group was used to establish a baseline measurement of the types of responses to potentially sensitive social issues that subjects were willing to give in isolation. These were then used in fictitious group settings to construct treatments intended to generate social pressure in the opposite direction of the baseline measurements.

1. The Behavioral Task

To relate the experimental design to the preceding discussion, a behavioral task had to be chosen that would tap subjects' levels of adherence to egalitarian norms. In the United States, as in many other multicultural societies, it is without controversy to claim that in most public social settings, norms of appropriate social behavior dictate compliance with a norm of individual and group equality, without respect to race, ethnicity, religion, gender, and so forth. That is, individuals' true attitudes with respect to, say, religious equality, are expected to vary—sometimes greatly—in reality, but the range of appropriate public expression of religious attitudes lies wholly on the equality side of the scale. With respect to specific identifiable groups, however, tolerance of dissent from the norm of equality can vary at any particular time given contemporaneous social circumstances. Therefore, in order to examine differences between subjects who are willing to express an inegalitarian attitude and those who are not, a target group had to be chosen that would elicit a relatively high level of variation in subject responses. Based on a variety of polling data, as well as a social climate at the time the study was conducted in which the threat of violence by Islamist groups loomed large in the Western public conscience, expressed attitudes about Muslims and Islam by non-Muslims was chosen as the behavioral task of interest for examining social pressure.¹⁾

Table 1 presents the questionnaire items that were used as behavioral measures to

test the various hypotheses put forth above. For each question, a determination was made as to what constituted the egalitarian and inegalitarian responses, and these decisions are indicated in the latter columns of the table. The number of response options available to subjects on the questionnaire is also indicated. Further, each question is given an identification number (first column). In much of the discussion that follows, it will be more efficient to refer to questionnaire items by their identification number rather than their full question wording.

Table 1. Summary of Questions About Muslims and Islam Used to Examine the Effect of Social Pressure

ID	Question Wording	Response Scale	Egalitarian Response	Inegalitarian Response
Q1	Would you be nervous if a Muslim man were on the same flight as you?	2-point	No	Yes
Q2	Would you mind living in a neighborhood where half of your neighbors are Muslim?	2-point	No	Yes
Q3	Do you favor special security checks at airports for Muslims?	4-point	Strongly oppose/ Oppose	Strongly favor/ Favor
Q4	Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?	4-point	A little/ Not at all	A moderate amount/ A lot
Q5	In your best estimation, what proportion of Muslims in the world are anti-American?	3-point	Some or just a few	Most or almost all/About half
Q6	Do people have the right to keep Muslim organizations out of their neighborhoods?	2-point	No	Yes

Table continued on next page.

1) The incentive for subjects to participate in the study was extra course credit for one of their political science courses, and therefore, in the name of fairness, all students in a given course were offered the opportunity to participate. That is, Muslim students were not excluded from the recruitment pool. However, a survey question asked students for their religious affiliations. In all of the analyses that follow, self-identified Muslims are excluded.

Q7	Would you mind if you yourself or your children were to have a Muslim teacher at school?	2-point	No	Yes
Q8	Do you think that most, many, only some, or very few Muslims living in the United States today support the use of violence by Muslims against non-Muslims?	4-point	Only some/ Very few	Most/Many
Q9	How worried are you, if at all, about radicals within the U.S. Muslim community?	3-point	Not worried at all	Very worried/ A little worried
Q10	Would you mind if your children were to attend a predominantly Muslim school?	2-point	No	Yes
Q11	In general, do you think that Islam is more likely than other religions to encourage violence among its believers?	2-point	No	Yes
Q12	Do you think that most, many, only some, or very few Muslims living in the United States today are peaceable and do not condone violence?	4-point	Most/Many	Only some/ Very few
Q13	Would you mind having a Muslim supervisor at work?	2-point	No	Yes
Q14	Would you be nervous if a Muslim woman were on the same flight as you?	2-point	No	Yes
Q15	Would you mind if a close relative who is a male married a Muslim?	2-point	No	Yes
Q16	Would you mind if a close relative who is a female married a Muslim?	2-point	No	Yes
Q17	Do you think that most, many, only some, or very few Muslims living in the United States today support the goals of Al Qaeda and the Islamic fundamentalists?	4-point	Only some/ Very few	Most/Many
Q18	Do you favor laws that prohibit discrimination against Muslims?	4-point	Strongly favor/ Favor	Strongly oppose/ Oppose

Table continued on next page.

Q19	Based on the events of September 11th, 2001, would you favor or oppose the U.S. government tightening immigration laws to restrict the number of immigrants from Arab or Muslim countries entering the United States?	4-point	Strongly favor/ Favor	Strongly oppose/ Oppose
Q20	Should someone who says that terrorist attacks against America are the fault of how America behaves in the world be allowed to teach in a public school?	2-point	No	Yes

Note: ID numbers are used to refer to particular questions in the text. Combined response options in the egalitarian and inegalitarian categories are separated by slashes.

2. Participants

Subjects were recruited from undergraduate political science courses at the University of Illinois at Urbana-Champaign in exchange for extra credit. Subjects were recruited under the auspices of taking part in a study on “group and individual decision making.” Summary statistics for the participants included in the analyses are presented in the Supplemental Appendix.

3. Design Overview

In the study, a control group of subjects came to the laboratory and answered a questionnaire that tapped, among other items, attitudes about Islam and Muslims. These control measures were then used to construct a treatment intended to generate social pressure in the opposite direction of that established in the baseline setting (this is similar to the method used by Sherif 1935, 1936; see also Jacobs and Campbell 1961). This social pressure treatment was administered via a contrived group setting in which subjects in the treatment group were provided information

about the distribution of opinion among a group of other participants whom they would meet in a face-to-face setting at a later time.

1) Procedure for the Control Group

Up to 20 subjects at a time gathered in a waiting room adjoined to a set of individual, private, enclosed computer terminals. Subjects arrived individually, typically over a period of 10 to 20 minutes. After filling out preliminary paperwork, subjects were told that they would be participants in a broader research study on group and individual decision making, and that today, they would simply take part in an individual survey. Subjects were then assigned to a computer terminal, and followed a set of instructions as the software led them through a questionnaire.

To simply gather a distribution of responses to the questionnaire, it was not strictly necessary to have subjects gather together in a common waiting room or to go through the computer-based survey in an enclosed, private computer terminal. But these steps were taken in an attempt to eliminate any confounding effects of the experimental context itself. Given that the primary measures of interest in the study are subject responses to questions tapping sensitive social issues, it seems reasonable to expect that certain features of the laboratory setting may come to bear on the extent to which individuals feel confident about the anonymity of their responses. Specifically, it is likely that the social nature of the way the subjects gather and the privacy of the physical setting in which they answer the questionnaire could each impact the degree of general social desirability bias exhibited in the survey responses. The goal of the control setting was not to eliminate biases in responses stemming from social desirability effects, but rather to ensure that such biases can be excluded as a competing explanation for any differences witnessed between the treatment and control groups.

2) Procedure for the Treatment Group

Overview After being given information about the supposed distribution of attitudes on several public opinion items among a subset of other study participants, subjects answered a computer-based questionnaire tapping those same attitudinal measures, among others. A ruse was employed to convince subjects that their own responses to the questionnaire would eventually be made public to that same group of other participants, following which time they would engage in a face-to-face task with them. Because the manipulation relied on a deception, following the questionnaire subjects were debriefed about it and queried as to whether the deception was convincing.

Preliminaries Up to 10 subjects at a time gathered in the same waiting room that was used in the control condition. Again, subjects arrived individually, typically over a period of 10 to 20 minutes. As subjects filled out preliminary paperwork, the researcher informed them that another group of subjects was taking part in a study in a nearby room, and that the progress of that group would have to be monitored. This other group of participants did not in fact exist, but to complete the ruse, the researcher periodically left the room to supposedly check on them. Specifically, the researcher would apologize and leave the room for a period of one to two minutes. Upon returning to the room, the researcher would go to a computer in the waiting room and look through a spreadsheet full of seemingly random numbers in full view of the participants. The purpose of each of these steps was to make the deception more convincing. This process was then repeated every several minutes until the agreed-upon start time of the study had passed and each subject had experienced the researcher leaving the room and returning at least once. At that point, the researcher addressed the entire group of subjects, instructing them to file into their assigned computer terminals, close the door, and await further individual instructions while the researcher confirmed the status of the supposed

other group's progress. Specific instructions are provided in the Supplemental Appendix.

Individual Instructions 1 As the subjects filed into their assigned computer terminals and closed the doors, the researcher left the room for a period of one to two minutes to supposedly check on the other group of participants. Upon returning, the researcher walked into each computer terminal and gave a set of supposedly individualized instructions. In fact, however, the instructions recited to each subject were identical. Specifically, the researcher walked into each computer terminal carrying a file folder containing four sheets of paper with printed material on them. The researcher informed subjects that part of the purpose of the study was to analyze group decision-making situations when people are allowed to learn about their fellow group members prior to engaging in a face-to-face task. To that end, the subjects were told that they were assigned to a group composed of four people from the other room, and that the printed material in the file folder represented the responses of these four people to six randomly chosen questions from a questionnaire that they had answered. To eliminate possible suspicion about the deception at this point in the study, the researcher informed subjects that the participants today were assigned two different roles, and that the difference between the people who gathered with the subject and the people in the other room was that the latter did not know about the group component prior to answering the questionnaire, and that this difference would be used to test something. The researcher instructed subjects on how to read the printouts of others' responses, and asked them to look them over for a few minutes and await further instructions. The printed materials given to subjects are presented in Figure 1 and the specific instructions recited are presented in the Supplemental Appendix. The researcher then left the private computer terminal, closed the door, and in succession gave the same set of instructions to the remaining subjects.

Individual Instructions 2 After each subject went through the procedure de-

SURVEY RESULTS	SEPTEMBER 28, 2011	SESSION 2	PARTICIPANT # 3	
Do people have the right to keep Muslim organizations out of their neighborhoods?	Yes X	No		
Do you approve or disapprove of the way that President Obama is handling the situation in Afghanistan?	Strongly Approve	Approve X	Disapprove	Strongly Disapprove
Do you favor or oppose the idea that homosexuals should be allowed to serve openly in the United States military?	Strongly Favor	Favor	Oppose	Strongly Oppose X
Would you be nervous if a Muslim man were on the same flight as you?	Yes X	No		
Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?	A lot X	A moderate amount	A little	Not at all
In your best estimation, what proportion of Muslims in the world are anti-American?	Most/Almost All X	About Half	Some/Just a Few	

SURVEY RESULTS	SEPTEMBER 28, 2011	SESSION 2	PARTICIPANT # 4	
Do people have the right to keep Muslim organizations out of their neighborhoods?	Yes X	No		
Do you approve or disapprove of the way that President Obama is handling the situation in Afghanistan?	Strongly Approve	Approve X	Disapprove	Strongly Disapprove
Do you favor or oppose the idea that homosexuals should be allowed to serve openly in the United States military?	Strongly Favor	Favor	Oppose X	Strongly Oppose
Would you be nervous if a Muslim man were on the same flight as you?	Yes X	No		
Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?	A lot X	A moderate amount	A little	Not at all
In your best estimation, what proportion of Muslims in the world are anti-American?	Most/Almost All X	About Half	Some/Just a Few	

Figure 1. Example Printouts Given to Subjects During the Looming Social Pressure Experiment

Figure continued on next page.

SURVEY RESULTS	SEPTEMBER 28, 2011	SESSION 2	PARTICIPANT # 6		
Do people have the right to keep Muslim organizations out of their neighborhoods?	Yes X	No			
Do you approve or disapprove of the way that President Obama is handling the situation in Afghanistan?	Strongly Approve X	Approve	Disapprove X	Strongly Disapprove	
Do you favor or oppose the idea that homosexuals should be allowed to serve openly in the United States military?	Strongly Favor	Favor	Oppose X	Strongly Oppose	
Would you be nervous if a Muslim man were on the same flight as you?	Yes X	No			
Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?	A lot	A moderate amount X	A little	Not at all	
In your best estimation, what proportion of Muslims in the world are anti-American?	Most/Almost All	About Half X	Some/Just a Few		

SURVEY RESULTS	SEPTEMBER 28, 2011	SESSION 2	PARTICIPANT # 9		
Do people have the right to keep Muslim organizations out of their neighborhoods?	Yes X	No			
Do you approve or disapprove of the way that President Obama is handling the situation in Afghanistan?	Strongly Approve X	Approve	Disapprove	Strongly Disapprove	
Do you favor or oppose the idea that homosexuals should be allowed to serve openly in the United States military?	Strongly Favor	Favor	Oppose	Strongly Oppose X	
Would you be nervous if a Muslim man were on the same flight as you?	Yes X	No			
Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?	A lot	A moderate amount X	A little	Not at all	
In your best estimation, what proportion of Muslims in the world are anti-American?	Most/Almost All	About Half X	Some/Just a Few		

Note: Each panel is one page of a printout given to experimental subjects that supposedly represented the distribution of opinion among fellow group members across six randomly chosen questions. Materials given to subjects were identical across and within study days, with the exception that the date on the top of each page of the printout always reflected the current date. Materials were contained in a file folder with “# 11” written on the outside (see instructions in the Supplemental Appendix for details).

Figure 1. Example Printouts Given to Subjects During the Looming Social Pressure Experiment (continued from previous page)

scribed above, the researcher returned and informed them that they would now go through the same questionnaire answered by their fellow group members. They were also informed that following the questionnaire, they would have a brief break before the group decision-making task, at which time the researcher would make a similar printout about them and present it to their fellow group members before the initial face-to-face meeting. The group decision-making task, they were told, would be very brief, and then their participation in the study would be over. Then at their individual computer terminals, subjects answered the questionnaire, received a debriefing about the deception employed in the study, and then answered questions about whether the deception was successful.

IV. Results

Comparing the printed materials in Figure 1 to Table 1, we see that four of the six questions of interest were used to generate the social pressure treatment. For the question of whether subjects would be nervous if a Muslim man were on the same flight as them (Q1), the supposed distribution of opinion among the four group members was unanimously "Yes"; for the question of whether subjects would worry about having a Muslim as a neighbor (Q4), the distribution was two answering "A lot" and two answering "A moderate amount"; for the question of what proportion of Muslims in the world are anti-American (Q5), the distribution was one answering "Most/Almost all" and three answering "About half"; and for the question of whether people have the right to keep Muslim organizations out of their neighborhoods (Q6), the distribution was unanimously "Yes." All of the responses of future group members, therefore, fall on the inequalitarian side of the scale (see Table 1). In

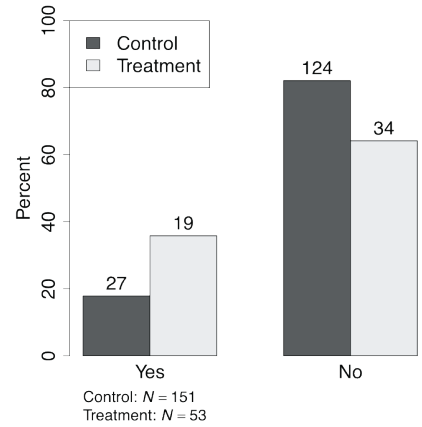
the discussion that follows, these questions will be referred to as direct questions, given that they are the questions for which direct social pressure was exerted. The remaining questions (Q2-Q3 and Q7-Q18) will be referred to as the indirect social pressure questions.

To get an initial sense of the distribution of responses in the treatment and control groups, Figure 2 presents barplots of the questions drawn directly from the materials given to subjects. For each question with more than two response options, the egalitarian and inegalitarian response categories are each collapsed. The first point to note is that the deliberate construction of a treatment to exert social pressure in the opposite direction of the distribution of opinion established in a control setting necessarily dictated that the treatment distribution would be skewed toward inegalitarianism. Comparing the heights of the dark bars within each panel, it is clear that subjects in isolation show an overwhelming tendency to choose the egalitarian response option on these particular questions. Given the large body of research on Americans' racial attitudes, this result is not at all surprising. For question Q4, the difference between the set of control subjects who gave the egalitarian and inegalitarian responses was about 93 percentage points.

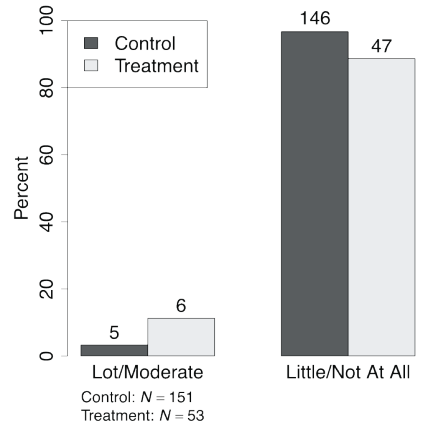
The second thing to notice is that the percentage point difference between the subset of subjects giving the egalitarian and inegalitarian responses was dissipated across the board in the treatment group. For each question, when informed that the attitudes of their fellow group members were unanimously on the inegalitarian side of the response scale, and under the expectation that their own responses would be made public to a group of peers, the percentage of subjects willing to give the inegalitarian response increased by about 18 percentage points for Q1, by about 1 percentage point for Q4, by about 13 percentage points for Q5, and by about 0.04 percentage points for Q6. The directional effects clearly exhibit initial support for the hypothesis, but it is not yet obvious that these differences are significant.

Focusing just on these direct questions, Table 2 presents χ^2 difference-of-propor-

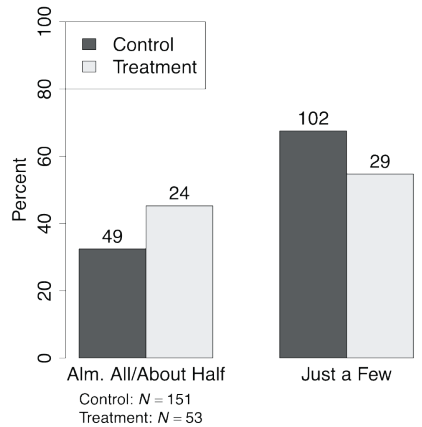
Q1: Would you be nervous if a Muslim man were on the same flight as you?



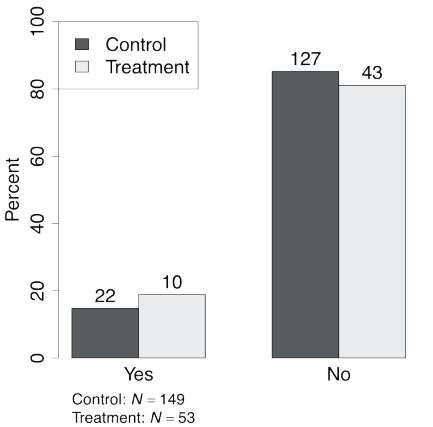
Q4: Would you worry a lot, a moderate amount, a little, or not at all about having a Muslim as a neighbor?



Q5: In your best estimation, what proportion of Muslims in the world are anti-American?



Q6: Do people have the right to keep Muslim organizations out of their neighborhoods?



Note: For response choices identified as egalitarian and inegalitarian, refer to Table 1. Combined response options in the egalitarian and inegalitarian categories are separated by slashes. Bar heights are percentages within categories. Numbers above bars are the actual number of respondents in each category.

Figure 2. Distribution of Egalitarian and Inegalitarian Responses in the Looming Social Pressure Experiment for Questions Drawn Directly from Materials

tion tests for the proportion of subjects giving the inegalitarian response between the treatment and control groups. Also presented are Fisher's exact tests of inde-

Table 2. Treatment-Control Differences in the Looming Social Pressure Experiment for Questions Drawn Directly from Experimental Materials

ID	Control Inegalitarian Proportion	Treatment Inegalitarian Proportion	$\chi^2(1)$	Asymptotic <i>p</i> -value	Fisher's Exact <i>p</i> -value
Q1	0.182	0.358	5.893	0.008	0.009
Q4	0.034	0.113	3.347	0.034	0.039
Q5	0.324	0.453	2.272	0.066	0.067
Q6	0.151	0.189	0.182	0.335	0.328

Note: Second and third columns are the proportions of respondents in the control and treatment conditions, respectively, that gave the response identified as inegalitarian (see Table 1). Null hypothesis for Fisher's exact test is that the number of people who gave the inegalitarian response is unrelated to whether they were in the treatment or control group. The alternative hypothesis is that the number of people in the treatment group who gave the inegalitarian response is greater than the number in the control group. Asymptotic *p*-value is based on the $\chi^2(1)$ approximation of the hypergeometric null distribution (Fisher 1922) using Yates's continuity correction (Yates 1934).

pendence for the contingency table in which subjects are classified according to whether they were in the treatment or control group and whether they gave the inegalitarian response. For both tests, the null hypothesis of no relationship was put in contrast to the alternative that the proportion or number in the treatment group was greater than the proportion or number in the control group. The χ^2 *p*-values and the exact *p*-values match quite closely, and they tell a story that comports with the illustration in Figure 2.

The two questions that seem to elicit the largest social pressure effect tap a general sense of concern among the subjects about being in close proximity to Muslims. Upon reflection, this sense of concern forms a qualitative similarity between Q1 and Q4 that also provides a qualitative difference with the other questions. When asked about the level of nervousness or worry about being on a flight with a Muslim or having a Muslim as a neighbor, subjects are more susceptible to the power of suggestion from their peers than with respect to other types of questions. For both of these questions, the difference in proportion of subjects giving the inegalitarian

response between the treatment and control groups achieves a conventional level of statistical significance, and the 17-percentage-point difference is particularly pronounced in the case of Q1. At least one plausible interpretation of these results is that a large proportion of non-Muslims carry some positive level of concern about being in close proximity to Muslims, but they typically allow that concern to remain latent if asked. Upon learning that a group of non-Muslims peers are willing to express openly their concern in these situations, those individuals receive a cue that it is permitted, if not necessary, to allow those concerns to come out of latency to the forefront. Notice that this interpretation is not stating that individuals are being compelled to express an attitude that they do not believe. Rather, it is simply stating that in the absence of a defined group opinion, the overwhelming tendency is for individuals to keep those concerns on the backburner. And it is reasonable to believe that at least one reason for this is the perception of an appropriate social response.

Q5, tapping subjects' perceptions of the pervasiveness of Muslim anti-Americanism, is qualitatively different in substance from Q1 and Q4, but the attitudinal result is similar. The difference in proportion giving the inegalitarian response between treatment and control groups achieves marginal statistical significance in this case. And because this question is closer to a statement of factual knowledge than questions tapping personal concern, it is difficult to provide the same interpretation. To the extent that individuals encounter this question having not considered it before, it is perhaps the case that the unanimously inegalitarian responses of fellow group members provide information about the correct answer to the question. To the extent that individuals do have well-formed attitudes with respect to this question, movement away from these attitudes must necessarily represent public compliance with views that they do not truly hold. But because it is likely that on-the-fly opinion formation and alteration of true attitudes are both occurring, teasing out the degree of simple learning and the degree of preference falsification is

simply not possible with the data at hand. What we can say is that there is strong evidence of a public compliance effect, regardless of the underlying mechanism producing it.

For Q6, the differences in proportions giving the inegalitarian response between the treatment and control groups does not come close to achieving a conventional level of statistical significance. Here the reason behind the result could plausibly reside in the fact that this particular question may be tapping less of a personal attitude about Muslims and more of a personal attitude about the civil rights of minority groups in general. Regardless of individuals' personal views of Muslims and Islam, the overwhelming majority is likely to recognize the rights of all groups to operate within the bounds of the law. In hindsight, the lack of a social pressure effect with respect to this question is not surprising, and in fact perhaps should have been expected.

As a point of comparison to the questions for which direct social pressure was exerted in the treatment, Table 3 presents χ^2 difference-of-proportion tests and Fisher's exact tests for the indirect questions, that is, the remaining questions on the survey tapping attitudes about Islam and Muslims for which no information about the distribution of peer responses was given. And the big takeaway point from this table is that very little in the way of differences between treatment and control groups is exhibited. Differences between treatment and control achieve marginal statistical significance for Q8 and Q9, which tap subjects' attitudes about militant Islam within the United States. Another question, Q11, also measures attitudes about radical Islam, and the differences between treatment and control in this case are approaching marginal significance. It is plausible to think that individuals who encounter a unanimous opinion from their peers on an issue to which they have not given much thought might use that information to make inferences about the distribution of opinion on related questions. Then public compliance to a perceived group attitude may operate on inferred group attitudes just as well as

Table 3. Treatment-Control Differences in the Looming Social Pressure Experiment for Questions Not Drawn Directly from Experimental Materials

ID	Control Inegalitarian Proportion	Treatment Inegalitarian Proportion	$\chi^2(1)$	Asymptotic <i>p</i> -value	Fisher's Exact <i>p</i> -value
Q2	0.426	0.453	0.033	0.428	0.427
Q3	0.223	0.264	0.175	0.338	0.333
Q7	0.082	0.113	0.167	0.341	0.584
Q8	0.061	0.152	2.758	0.048	0.054
Q9	0.709	0.830	2.370	0.062	0.059
Q10	0.581	0.623	0.134	0.357	0.359
Q11	0.333	0.434	1.303	0.127	0.127
Q12	0.054	0.019	0.468	0.247	0.259
Q13	0.095	0.057	0.320	0.714	0.876
Q14	0.047	0.075	0.178	0.337	0.322
Q15	0.182	0.154	0.065	0.600	0.747
Q16	0.320	0.358	0.118	0.365	0.362
Q17	0.047	0.038	0.000	0.500	0.607
Q18	0.524	0.415	1.433	0.884	0.546

Note: Second and third columns are the proportions of respondents in the control and treatment conditions, respectively, that gave the response identified as inegalitarian (see Table 1). Null hypothesis is that the number of people who give the inegalitarian response is unrelated to whether they were in the treatment or control group. Alternative hypothesis is that the number of people in the treatment group who give the inegalitarian response is greater than the number in the control group. Asymptotic *p*-value is based on the $\chi^2(1)$ approximation of the hypergeometric null distribution (Fisher 1922) using Yates's continuity correction (Yates 1934).

realized group attitudes. But if this is the case, then it is not clear why other questions tapping opinions about militant Islam in the U.S. (e.g., Q12 and Q17) do not show a similar effect. In fact, for no questions other than Q8, Q9, and Q11 do differences between treatment and control even come close to being statistically significant. The fact that each of these questions seems qualitatively to measure a similar subcategory of attitudes about Islam and Muslims is promising evidence for the case that something systematic may be left uncovered. But the overall picture is

that a form of indirect social pressure, in which subjects use available information to make inferences about social pressure in other areas, is not occurring in general.

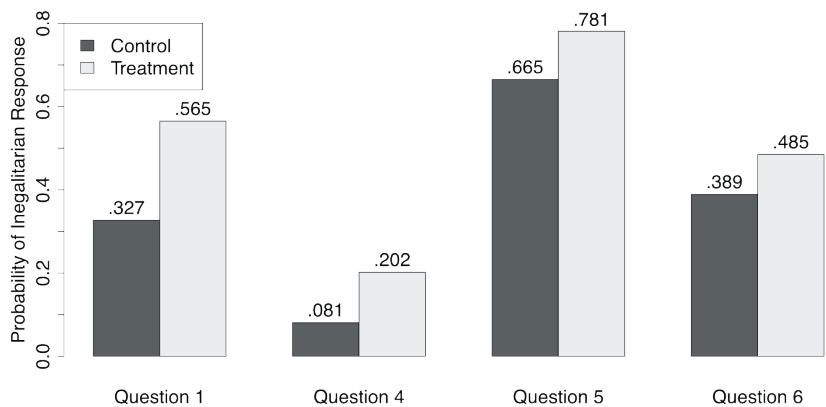
The analyses thus far have not accounted for individual-level characteristics beyond assignment to treatment or control groups. To control for characteristics of the individual, a series of logistic regressions were used to model the individual-level probability of providing an inegalitarian response. For each model in Table 4, the unit of analysis is the individual subject, and the binary dependent variable is 1 if the subject gave the inegalitarian response to the question identified in each column heading, and 0 otherwise. Dummy variables are used to capture race (1 = white), gender (1 = female), religion (1 = Christian), and partisan identification (1 = Republican), and a count variable is used to capture education (1 = first year through 4 = fourth year or greater). Interestingly, at this level of analysis, when controlling for race, gender, religion, partisan identification, and education, only Q1, the question tapping nervousness about being on a flight with a Muslim man, exhibits a treatment effect that achieves a conventional level of statistical significance. However, the estimated treatment effects for Q4 and Q5 come very close to a 90% confidence level (for Q4, $p \approx 0.107$, and for Q5, $p \approx 0.108$). As expected from the tests of independence and differences of proportions between treatment and control groups, the treatment effect for Q6 does not come close to achieving statistical significance. Among the components of the covariate profiles, the only substantial predictor of the likelihood of giving an inegalitarian response to these questions is partisan identification. Specifically, self-identified Republicans are significantly more likely to express nervousness about being on a flight with a Muslim man, to estimate that most or about half of Muslims in the world are anti-American, and to state that people have the right to keep Muslim organizations out of their neighborhoods. Additionally, a greater number of years in school is associated with a lower likelihood of estimating that most or about half of Muslims in the world are anti-American.

Table 4. Logistic Regression Models of Inegalitarian Responses on Questions Used for Direct Social Pressure

	Question 1 Estimate (s.e.)	Question 4 Estimate (s.e.)	Question 5 Estimate (s.e.)	Question 6 Estimate (s.e.)
Treatment	0.982* (0.385)	1.063 (0.660)	0.585 (0.362)	0.388 (0.456)
White	0.353 (0.465)	-0.405 (0.753)	-0.339 (0.397)	0.448 (0.597)
Female	0.162 (0.375)	0.325 (0.650)	0.159 (0.340)	0.441 (0.425)
Christian	-0.111 (0.424)	0.686 (0.850)	0.550 (0.389)	0.519 (0.553)
Republican	0.725# (0.421)	0.298 (0.766)	1.309*** (0.389)	1.227*** (0.446)
Education	-0.225 (0.198)	-0.570 (0.466)	-0.422* (0.182)	-0.003 (0.213)
Constant	-1.624 (0.611)	-2.767* (1.169)	-0.570 (0.526)	-3.080*** (0.794)
N	197	197	197	196
Log-likelihood	-98.187	-38.739	-115.031	-79.645

Note: Cell entries are logistic regression coefficient estimates; standard errors are in parentheses. Binary dependent variable is 1 if the respondent gave the inegalitarian response, and 0 otherwise. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; # $p < 0.10$.

Momentarily putting aside the lack of statistical significance for Q6, further interpretation is available through the calculation of predicted probabilities. Figure 3 provides a graphical representation of the predicted probability of giving the inegalitarian response for members of the treatment and control groups based on the estimates in Table 4. For each of these calculations, all other independent variables are held constant at 1, and therefore the baseline hypothetical subject under consideration is a white, female, Christian, Republican, college freshman. As expected, the most dramatic treatment-control probability difference is for Q1. A baseline subject in the control group has a probability of only 0.327 of openly expressing nervous-



Note: Predicted probabilities are for treatment and control groups for each model in Table 4. Calculations are for a baseline subject who is white, female, Christian, Republican, college freshman. Actual probabilities are presented above the bars.

Figure 3. Predicted Probabilities from Logistic Regression of Inegalitarian Responses on Questions Used for Direct Social Pressure

ness about being on the same flight as a Muslim man, while for a baseline subject in the treatment group this probability increases to 0.565 (an increase of roughly 0.24, from below the 0.5 threshold to above it). For Q4 and Q5, this increase is roughly 0.12, about half that of Q1. Interestingly, for Q5, tapping the estimation of anti-Americanism among Muslims, the baseline subject already exhibits a relatively high likelihood of providing the inegalitarian response (due largely to the strong impact of Republican party identification, but the treatment has the impact of increasing this likelihood even further. For Q4, both treatment and control subjects exhibit a relatively low likelihood of openly providing the inegalitarian response.

It may also be the case, however, that focusing on subject-level analyses for individual questions is somehow masking a more general process that might be revealed by pooling across responses to questions. To begin to examine the impact of combining responses across questions, Table 5 presents a set of logistic regressions where, again, the binary dependent variable is 1 if the subject changed her response, and 0 otherwise. But in this case, the unit of analysis is the subject's

Table 5. Logistic Regression Models of Inegalitarian Responses, Pooling Across Direct and Indirect Social Pressure Categories

	Indirect Questions		Direct Questions	
	Estimate	(s.e.)	Estimate	(s.e.)
Treatment	0.242	(0.166)	0.724***	(0.275)
White	-0.123	(0.176)	0.055	(0.318)
Female	-0.083	(0.156)	0.243	(0.262)
Christian	0.366*	(0.170)	0.322	(0.307)
Republican	0.653***	(0.174)	1.071***	(0.289)
Education	-0.106	(0.077)	-0.280*	(0.139)
Constant	-1.303***	(0.237)	-2.060***	(0.430)
Observations	2745		787	
Subjects	197		197	
Var(α_i)	0.582		0.582	
Log-likelihood	-1522.737		-362.458	

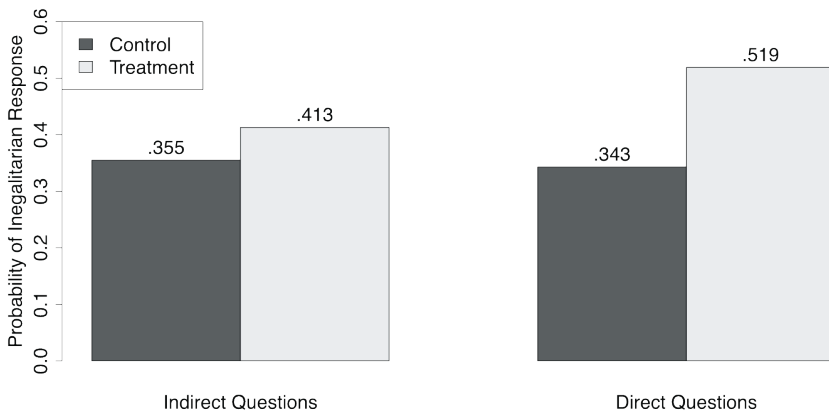
Note: Cell entries are logistic regression coefficient estimates, with standard errors in parentheses. Binary dependent variable is 1 if the respondent gave the inegalitarian response, and 0 otherwise. All models include subject-level random intercepts, α_i , where i indexes subjects. The unit of observation is the subject-question. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; # $p < 0.10$.

response to a particular question, so that each row of the data matrix is a unique subject-question combination. To account for the hierarchical nature of the data, subject-level intercepts are assumed to be drawn from a normal distribution and are allowed to vary randomly. The first model pools only across indirect questions and the second pools only across direct questions.

Because the directional relationships in the raw data for each question of interest here all point in the direction of support for the hypothesis, it is plausible to think that through borrowing strength across indirect questions and direct questions, the estimated relationship for the indirect questions may become more substantial, and that those for the direct questions may become less substantial (due to the lack of significance for the treatment-control differences with respect to Q6). But neither

of these are the case, and the relationships borne out by pooling are consistent with the previous discussions. First and foremost, the treatment effect in the case of the direct questions is to substantially increase the probability of expressing the inegalitarian response, while in the case of indirect questions the treatment has no estimated effect. In the case of the indirect questions, the only significant predictors of providing the inegalitarian response are subject self-identification as Christian and Republican. Both the condition of being a Christian and the condition of being a Republican (as opposed to not) are expected to increase the likelihood of providing an inegalitarian response to an indirect question. Moving to the case of direct questions, the impact of partisan identification as Republican operates in the same direction and becomes even more substantively meaningful. The impact of being a Christian, however, dissipates, while the number of years in school becomes significantly associated with a decrease in the likelihood of giving inegalitarian responses.

Once again, predicted probabilities help with interpretation of the findings. Figure 4 provides a graphical representation of the fixed-effects estimates from the



Note: Predicted probabilities are for treatment and control groups for each model in Table 5. Calculations are for a baseline subject who is a white, female, Christian, Republican, college freshman. Actual probabilities are presented above the bars.

Figure 4. Predicted Probabilities from Logistic Regression of Inegalitarian Responses, Pooling Across Direct and Indirect Social Pressure Categories

model in Table 5,²⁾ and the form of the characteristic subject is again assumed by setting the values of all covariates to 1. We find that the estimated increase in the probability of providing an inequalitarian response when moving from control to treatment is substantial (about 0.18). The condition of being treated with information about an inequalitarian response distribution among peers has the effect of moving the likelihood of providing an inequalitarian response from below to above the 0.5 threshold.

Combining the data into unique subject-question observations is not the only means of pooling information across questions, however. Using the subject as the unit of analysis, we may also consider counts and proportions of inequalitarian responses given over the course of the study. To that end, Table 6 presents a series of negative binomial regressions of the number of inequalitarian responses given by subjects,³⁾ alongside a series of ordinary least squares regressions of the proportion of responses that are inequalitarian. Data are presented for all questions of interest considered together, and for direct and indirect questions considered separately.

For the negative binomial specifications, a similar story emerges, but here it will once again prove useful to calculate predicted probabilities. Figure 5 presents predicted probabilities for each of the negative binomial specifications in Table 6, and for which the hypothetical subject under consideration is again assumed by setting

2) Estimates for the random components, α_i , are set aside for now. Addition of the subject-level random intercepts can be considered as deviations away from the global constant presented in Table 5.

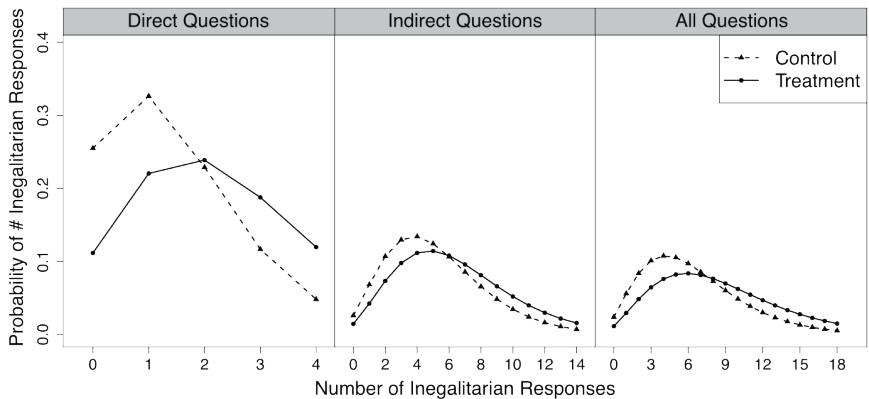
3) The negative binomial distribution was chosen as an appropriate conditional model by first modeling the counts of inequalitarian responses using a Poisson regression. A likelihood ratio test indicated that the addition of the estimated variance parameter of the negative binomial specification provided a significantly better fit to the data for the model using all questions and the model using only the indirect questions. This result provides strong evidence for overdispersion, indicating that the Poisson specification is inappropriate. In any case, the negative binomial model cannot be considered strictly appropriate for these data, as counts following a negative binomial process theoretically do not have an upper bound, whereas the counts presented here are bounded by the number of survey questions that respondents were asked. Nonetheless, the negative binomial model can still provide a useful approximation.

Table 6. Negative Binomial and Ordinary Least Squares Regressions of Counts and Proportions of Inegalitarian Responses

	All Questions		Indirect Questions		Direct Questions	
	Negative Binomial	OLS	Negative Binomial	OLS	Negative Binomial	OLS
	Estimate (s.e.)	Estimate (s.e.)	Estimate (s.e.)	Estimate (s.e.)	Estimate (s.e.)	Estimate (s.e.)
Treatment	0.301* (0.126)	0.081* (0.033)	0.208# (0.121)	0.060# (0.034)	0.514** (0.181)	0.108** (0.040)
White	-0.042 (0.135)	-0.013 (0.034)	-0.073 (0.129)	-0.019 (0.034)	0.070 (0.225)	0.009 (0.041)
Female	-0.093 (0.116)	-0.021 (0.030)	-0.092 (0.110)	-0.023 (0.030)	0.148 (0.177)	0.025 (0.037)
Christian	0.270* (0.125)	0.064* (0.031)	0.252* (0.121)	0.063* (0.032)	0.301 (0.221)	0.044 (0.039)
Republican	0.498*** (0.125)	0.138*** (0.033)	0.443*** (0.119)	0.130*** (0.034)	0.687*** (0.189)	0.153*** (0.042)
Education	-0.097 (0.059)	-0.020 (0.015)	-0.095# (0.056)	-0.021 (0.015)	-0.177# (0.096)	-0.032# (0.018)
Constant	1.330*** (0.180)	0.209*** (0.045)	1.226*** (0.171)	0.242*** (0.045)	-0.650* (0.307)	0.143* (0.056)
θ	3.690 (0.747)		5.268 (1.399)		10.308 (14.429)	
N	184	184	184	185	196	196
Log-likelihood	-447.536		-415.153		-221.884	
R2		0.168		0.144		0.148

Note: Cell entries in the negative binomial columns are negative binomial regression coefficient estimates, where the dependent variable is the number of egalitarian responses given, θ is the estimated conditional variance of the response variable. Cell entries in the OLS columns are ordinary least squares coefficient estimates. Standard errors are in parentheses. The unit of observation is the subject. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; # $p < 0.10$.

all covariates equal to one. For each of the ways of subsetting the data, members of the control group have a higher probability of espousing a low number of inegalitarian responses. For instance, out of the four direct questions, the members of the control group have a higher probability of expressing no or only one in-



Note: Predicted probabilities are for treatment and control groups for each negative binomial model in Table 6. Calculations are for a baseline subject who is a white, female, Christian, Republican, college freshman. Lines connecting plotting symbols are presented as a visual aid.

Figure 5. Predicted Probabilities from Negative Binomial Regressions of Counts of Inegalitarian Responses

egalitarian response. The members of the treatment group, however, have a higher probability of expressing two, three, or four inegalitarian responses. In the model for indirect questions only, out of the fourteen possible responses, members of the control group have a higher probability of expressing zero through five inegalitarian responses, while members of the treatment group have a higher probability of giving seven through 14. In the model that uses all questions combined, out of 18 possible responses, the probability of expressing zero through seven inegalitarian responses is higher for members of the control group than for members of the treatment group, whereas the probability of expressing eight through 18 inegalitarian responses is higher for members of the treatment group than for members of the control group. In addition to there being a generally higher probability of expressing a relatively large number of inegalitarian responses in the treatment group, for each model it is also true that the peak probability occurs at a lower number of inegalitarian responses in the control group than in the treatment group.

V. Discussion and Conclusion

The results of the looming social pressure study have been examined in a variety of ways, all of which point to at least weak support for the prediction. As we know from a plethora of previous public opinion research, when citizens in advanced democracies are asked to express their attitudes on possibly sensitive social issues, the attitude distribution is typically skewed toward whatever happens to be the so-called politically correct response. In the case of questions tapping a sense of equality between racial, ethnic, religious, gender, or other groups, the politically correct response in modern times is certainly in favor of egalitarianism under most circumstances. This result is exactly what was seen with respect to the control group. However, when individuals are able to learn about the distribution of opinions in a peer group that they will later have to meet for an unspecified face-to-face decision-making task, and when that distribution is skewed in favor of inequality, the results presented here suggest that they can be pulled away from a general egalitarian position. These data do not allow us to say whether individuals in the treatment group are actually generally in favor of a more egalitarian position and some significant number are falsifying their preferences to publicly comply with the group opinion; or whether members of the control group are actually generally in favor of a more inequality position, and some significant number are altering their responses because of typical social desirability effects; or whether a significant proportion of subjects simply do not have a well formed attitude with respect to many of the questions in the survey, and they are looking to whatever information is available to form an opinion on the fly. Without knowing the source of the tendency toward public compliance, however, the results are clear that a significant amount of public compliance is in fact occurring.

Future research will attempt to parse the specific mechanism of conformity un-

derlying expressed attitude changes by using a within-subjects design that taps differences in attitude expression between private settings and opposed social pressure settings. While the potential limitation of the current approach is that the underlying mechanism remains hidden, the potential limitation of the within-subjects approach is that asking the same questions to subjects multiple times may generate a need for consistency across instances of attitude expression that acts as a counterforce for a mechanism of mere public compliance, if that represents the true underlying dynamic. In any case, to determine whether public compliance with respect to egalitarian attitudes operates more like learning or attitude change on the one hand, or temporary conformity on the other, we will have to examine the full body of evidence across multiple studies. For now, the results suggest that in an application regarding attitudes about inter-group relations between salient social groups in the United States, social pressure toward inequality can indeed lead individuals to abandon egalitarian attitudes supported by broad social pressure in favor of inequality attitudes among a proximate peer group.

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■ **David J. Hendry** 플로리다주립대학교 경영학과를 졸업하고 일리노이대학교 어바나-섐페인에서 정치학 박사 학위를 받았다. 현재 런던 정치대경제대학교 교수로 재직중이다. 주요 논문으로는 "Self Interest, Beliefs, and Policy Opinions: Understanding the Economic Source of Immigration Policy Preferences"(*Political Research Quarterly*, 2017, 공저)와 "Reassessing Schoenfeld Residual Tests of Proportional Hazards in Political Science Event History Analyses"(*American Journal of Political Science*, 2015, 공저) 등이 있다.

[Supplemental Appendix]

Summary Statistics from Experiments

The following table presents summary statistics for the experimental participants.

Table SA.1: Summary Frequency Distributions of Categorical Variables for Subjects in the Looming Social Pressure Experiment

		Control		Treatment	
		Frequency	Percentage	Frequency	Percentage
Gender	Female	40	27.21	25	47.17
	Male	107	72.79	28	52.83
Race	White	112	75.68	40	75.47
	Black	8	5.41	3	5.66
	Asian	19	12.84	5	9.43
	Other	9	6.08	5	9.43
Religion	Christian	99	67.81	38	71.70
	Non-Christian	47	32.19	15	28.30
Education	1st Year	60	40.54	36	67.93
	2nd Year	48	32.43	10	18.87
	3rd Year	27	18.24	4	7.55
	4th Year +	13	8.78	3	5.66
Party ID	Democrat	56	38.10	29	54.72
	Independent	26	17.69	8	15.09
	Republican	41	27.89	9	16.98
	Other	24	16.33	7	13.21
Ideology	Liberal	49	33.11	18	33.96
	Moderate	55	37.16	20	37.74
	Conservative	37	25.00	10	18.87
	Other	7	4.73	5	9.43

Experimental Instructions

Here I present the instructions read to participants and the laboratory materials used in the experiments.

After all participants in a given session had gathered in a common waiting room and

filled out a consent form to participate, the researcher recited to the group the following set of instructions:

Today you are participating in a study on group and individual decision making. The first part of the session will take place on these computer terminals. So you all can each choose an open terminal. You should go into the room and shut the door. I am going to come around to you one-by-one and give you instructions. It may take a few minutes for me to get to you. You may read or use the computer while you wait for me. But you will find that there is a web browser open and minimized on your computer. Don't touch that until I tell you to do so. So go ahead and choose a computer. What you are going to do has to be coordinated with the other group of participants, so I have to go make sure that everything is ready in the other room.

As the participants filed into the computer terminals and closed the door, the researcher left the room. After a period of one to two minutes, the researcher returned, walked into each terminal, one-by-one, and recited the following set of instructions:

Part of what we are interested in here today is what happens in group decision-making situations when people are able to learn a little bit about one another before they get together in groups to do something.

So there is a group of people in another room who have filled out a questionnaire with a series of public opinion questions. I have selected six of the questions at random, and I am going to give you a computer printout of four people's answers. You have been assigned to be in a group with these four people for something that happens later.

The people participating in the experiment today have been assigned different roles. For the purposes of this study, the only difference between the people in this room and the people in the other room is that the people in the other room filled out the questionnaire without knowing anything about the group part of the study. You have a different role in that you will be allowed to learn a little bit about

your fellow group members before you do anything for the study.

[Researcher points to the top page of a stack of computer printouts (refer to Figure 1).] On the printout, you'll see a survey question along with the set of responses, and then you'll see an X marked next to the other person's answers. Other people are identified only by a participant number in the upper right corner of the printout. And for this study, you are participant number 11.

So I will leave these printouts of the other people's answers with you for a couple of minutes. Just look them over and I'll be right back after I give these instructions to a few other people. [Researcher hands stack of printouts to subject, leaves the room, and closes the door.]

The researcher then left the room to give the instructions to another subject. After all subjects were given the above set of instructions, the researcher began again with the first subject, walked into each terminal, and recited the following set of instructions:

So now you are going to fill out the same questionnaire that your fellow group members already completed. When you're finished, there will be a brief break for you, and at that time I am going to give your fellow group members a similar printout about you. After that, you will go through a group task that takes just a few minutes.

The questionnaire has about 60 questions on it, and it generally takes people about 25 to 30 minutes to complete. You'll answer this, and then there will be a short break before the group task. And the software should provide all of the instructions that you need as you navigate through the questionnaire. Just read everything very carefully and go ahead and get started.

국문초록**사회적 압력이 여론에 미치는 순응 효과**

데이비드 헨드리

런던정치경제대학교

최근 수십 년 동안 사회적 규범의 눈에 띄는 변화는 사회 집단들의 관계를 정의하는 위계적 구조에 대한 지지로부터 평등주의적 가치를 지지하는 위계제에 대한 거부로의 움직임이다. 그러한 규범적 대체의 과정에서 개인들은 새롭게 사회적으로 적절한 태도를 가질 것을 기대되며 이는 사회적 상호작용을 통해 강화될 것으로 여겨진다. 이러한 과정은 광범위하게 이론화되고 논의되어왔으나 대부분의 규범의 강화가 주로 일상의 사적인 인간관계에서 발생하기 때문에 이를 경험적으로 조사하기에는 매우 어려운 측면이 있다. 본 논문은 동료집단과 근접한 집단을 설정하고 이들에 대한 설문 실험을 통해서 사회적 압력이 적어도 실험 참여자들에게 평등주의적 태도로부터 멀어지게 하고 동료집단의 의견에 가까워지게 한다는 것을 보여주고 있다.

주요어 순응효과, 사회적 규범, 사회적 압력, 평등주의, 정치적 가치